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Phase evolution of highly immiscible alloys under shear deformation: Kinetic pathways, steady states, and the lever-rule



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ABSTRACT

Phase evolution of dilute, highly immiscible, Cu alloys (Cu-Nb, Cu-V, and Cu-Ta) during low-temperature severe plastic deformation was investigated using large-scale molecular dynamics simulations. At low solute concentrations, each system maintained a FCC structure in steady state, but as the concentration was increased above a saturation limit (0.3 at.% for Ta, 1 at.% for Nb and 5 at.% for V), the system became two-phase, comprising co-existing FCC and amorphous phases. Unlike Cu-Nb and Cu-V, the amorphous phase in the Cu-Ta system showed strong solute partitioning. Increasing the solute concentration above a second phase boundary (8 at.% V, 9 at.% Nb, and 24 at.% Ta) led to complete amorphization. Throughout the two-phase region, the compositions of the FCC and amorphous phases remained nearly constant, thus following the lever rule. Initiating the systems either as a FCC homogeneous alloy, or with a BCC sphere embedded in a Cu matrix, had no effect on the steady state microstructure, implying uniqueness of the steady state under low-temperature shear deformation. Lastly, chemical order and phase partitioning in the amorphous Cu-Ta system under low-temperature shear is found remarkably similar to that in the equilibrium structure above the melting temperature.

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1. Introduction

Severe plastic deformation (SPD) of metal alloys during ball milling [1], accumulative roll bonding [2], or high pressure torsion [3] often leads to homogenization of alloy components, generally referred to as mechanical alloying [4]. Homogenization is even observed in immiscible alloys, such as Cu-Ag, Cu-Fe, Cu-Co and many others [1]. Various mechanisms for this homogenization process have been suggested, see for example refs [5–9,40], although a unified understanding of this process remains controversial. Some immiscible alloys, such as Cu-Ta [10], Cu-Nb [11], Cu-Mo [12,13], Ni-Ag [14] and Fe-Ag [41], on the other hand, fail to homogenize during SPD, even when shear strains are extremely high and shearing temperatures very low. Common to many of these alloys is a high heat of mixing, $\Delta H_m > 0.2$ eV/atom, at the equimolar solution, and it has been suggested from molecular dynamics [15] and molecular statics computer simulations [16] that

chemical interactions bias the relocation of atom during shearing, preventing complete homogenization. These past simulation studies, however, reported on short range ordering during shear in modestly immiscible alloys and did not reveal the two-phase co-existence found experimentally in systems with very high heats of mixing. It has also been suggested that the large difference in the strengths of the two immiscible phases might also contribute to the lack of homogenization [17]. An interesting observation regarding the shear-induced mixing of these highly immiscible alloy systems is that while they do not homogenize during extended ball milling or high pressure torsion (HPT), the length scale of the phase-separated phases decreases dramatically from their initial values, from several hundreds of nanometers or more, to some tens of nanometers [18,19]. For dilute Cu-Nb alloys, new studies [20] using HPT have further revealed that the same approximate microstructure evolves whether the initial state of the system is a two-phase alloy containing large Nb precipitates or a homogeneous solid solution. These results suggest that the Cu-Nb alloy self-selects a length scale during low-temperature shear, independent of its initial state. At present there are no detailed models explaining the behavior of these strongly immiscible alloys under low temperature shear deformation. Curiously, however, this behavior is

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reminiscent of low-temperature particle irradiation of strongly immiscible Cu alloys [21]. For example, irradiation of dilute Cu-W alloys [22] results in a large supersaturation of W in Cu, ~1.5 at.%, and narrow size distribution of W precipitates, both of which being independent of the initial state of the system. In this case the controlling mechanisms have been attributed to a dynamical competition between energetic (random) recoil mixing of atoms and solute agglomeration within local melt zones in energetic displacement cascades, i.e. thermal spikes. While these two irradiation mechanisms are clearly not involved in shearing, we will show that a corresponding dynamical competition occurs during low temperature shearing. The present paper thus explores self-organization behavior in highly immiscible Cu alloys during shear deformation using large-scale molecular dynamics simulations. Three alloys are selected for study, Cu-Nb, Cu-Ta, and Cu-V; while they all have large positive enthalpies of mixing, the atomic size mismatch and the relative elastic properties in these systems are quite different.

2. Computational methods

The simulations were performed using dilute Cu alloys as model systems, represented by embedded atom method type potentials [23] for Cu-Nb [24] and Cu-V [25] and a closely related angular dependent potential for Cu-Ta [26]. Table 1 lists properties of the alloy systems derived from these potentials. The MD code LAMMPS [27] and visualization using OVITO [28] were employed to run these simulations and help analyze the results. The samples in this study were created by two methods. For the first, a single spherical precipitate was introduced into the Cu matrix, selecting an arbitrary, low symmetry orientation. The cell was then relaxed by simulated annealing at 600 K at a hydrostatic pressure of 1 kbar for 100 psec, before bringing it to a temperature of 300 K at a pressure of 0 Pa. The annealing procedure was repeated 10 times to ensure stabilization. This procedure conserves precipitate shape while promoting interface relaxation. We note that at 600 K, diffusion is negligible, and thus the systems could stabilize during the short-term anneals without significant solute redistribution. In the second case, solute atoms were placed on randomly selected lattice sites in the Cu matrix and then relaxed by a similar annealing procedure. The two systems were then sheared cyclically using biaxial compression along the $\langle 100 \rangle$ (x,y,z) directions of the Cu matrix. In each cycle the sample was deformed to an engineering strain of 0.1 first in both the x and y directions, followed by similar strains in the y-z and z-x directions. This leads to an applied biaxial strain of 0.2 strain per compression and a total strain of 0.6 per cycle. All results are reported in terms of biaxial strain, ϵ , calculated using the trace of the imposed engineering strain tensor, which for biaxial compression is equal to the equivalent von Mises strain when neglecting elastic strains and small shape changes (see further details below). During the bi-axial compression, the third axis is maintained at zero stress leading to its subsequent elongation during the strain step. Thus following a full cycle, shape is nearly conserved and the system returns close to its initial dimensions. Since the strain is not completely isotropic, changes in the elongation along the free direction may lead to small changes in

shape. Prior to each compression cycle, therefore, strain rates were modified by up to 2% per orientation, relative to the nominal value, in order to preserve the shape of the computational cell at the end of each cycle. Simulations were performed using strain rates between $1 \times 10^8 \text{ s}^{-1}$ and $5 \times 10^9 \text{ s}^{-1}$, with no significant difference in the resulting steady states, but this will be discussed in more detail below. This insensitivity to strain rate indicates that there are no significant thermal contributions to the observed kinetics, although these are expected to be small in any case during deformation at 300 K. Results were also insensitive to changes in the total strain accumulated per compression, at least between 0.15 and 0.25. This is consistent with the fact that elastic strains per step are very small, accounting for less than 1% of the imposed strain. For strains less than 0.1 per step, elastic strains become more noticeable, especially in the initial stages of the simulations, reaching $\approx 10\%$ of the imposed strain, and leading to results that are sensitive to the strain per step. System sizes were varied between 10^5 and 10^8 atoms, depending on the alloy concentration. We verified that there were no significant effects of size on the steady state configurations, as characterized by the phase composition, energy and chemical order, although some qualification of this point will be presented in section 4. Systems were simulated until steady states were attained; this required total accumulated bi-axial strains of up to ~ 600 in some cases. It is noteworthy that strains of the order of 100's – 1000's are available experimentally using techniques such as ball-milling and HPT, and sometimes these high strains are required to achieve steady state. The strain required for steady state, however, depends on the length scale of the relevant microstructural features. Since our computational cell is quite small, the strain required for steady state tends also to be small. The small size, however, limits the materials problems we can investigate, as we will illustrate in the discussion section.

Structural identification of atoms was performed using the centro-symmetry parameter, which assigns low values to atoms with high crystalline symmetry [42]. Atoms within the amorphous phase were identified as those that have an average centro-symmetry parameter higher than 8, which was taken as an average over the atom in question and its 12 nearest neighbors. While centro-symmetry parameter is highly efficient in determining if an atom is located at a crystalline site, the local averaging ensured that only atoms embedded in non-crystalline environment were identified as amorphous, thus minimizing the counting of point defects.

3. Results

3.1. Dynamics of phase evolution during severe plastic deformation

We begin by providing an overview of the range of shear mixing behaviors in highly immiscible alloys for three specific alloys, $\text{Cu}_{94}\text{Nb}_6$, Cu_{94}V_6 , and $\text{Cu}_{86}\text{Ta}_{14}$. The initial conditions for these simulation events was a spherical BCC precipitate embedded in a FCC Cu matrix. Deformation begins with dislocations gliding in the Cu matrix, with no dislocations observed cutting through the precipitate, and virtually no co-deformation of the BCC particle taking place. This is seen in Fig. 1(a–c) where a snapshot of the dislocation

Table 1
Properties of solutes derived from EAM potentials. Heat of mixing (H_0) and atomic volume (V_0) provided at infinite dilution in FCC Copper. Atomic volume of copper is $11.81 \text{ \AA}^3$. Subscripts c and a refer to crystalline and amorphous states.

	G_{Cu} (GPa)	$G_c(5\%)$ (GPa)	$G_a(10\%)$ (GPa)	$V_0(\text{\AA}^3/\text{atom})$	$H_0(\text{eV}/\text{atom})$
Cu-Nb [24]	72	30	5	17.9	1.3
Cu-V [25]	72	25	17	14.0	0.75
Cu-Ta [26]	76	30	19	18.0	1.6

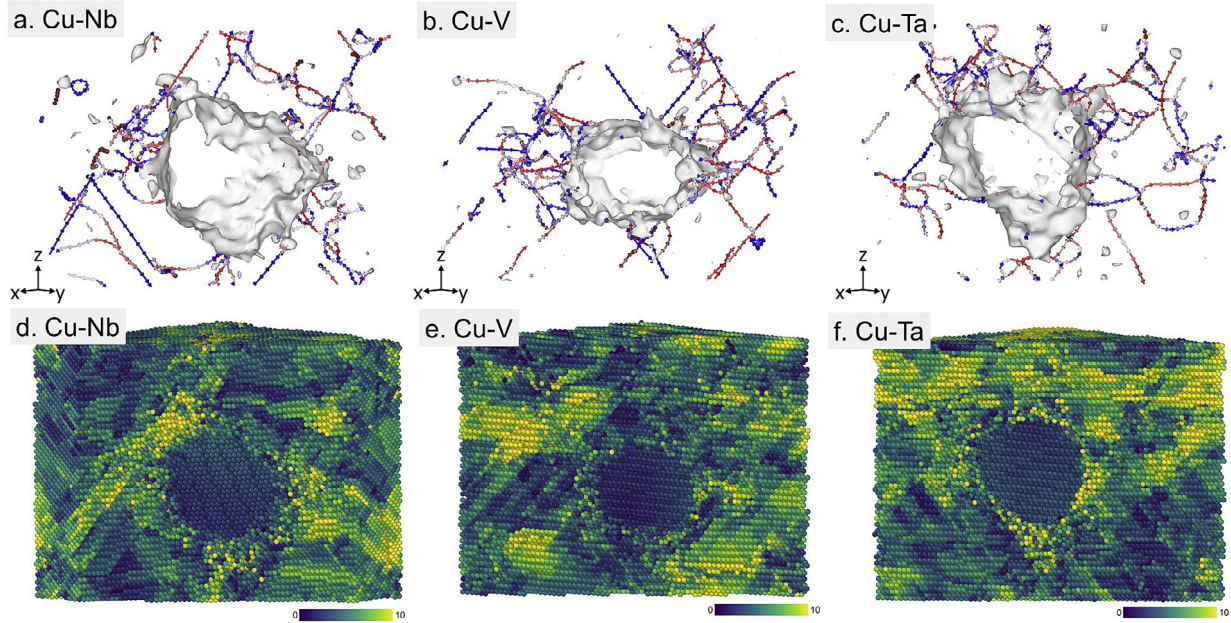


Fig. 1. Snapshot of a (110) cut through the simulation cell at a strain of ~40 for the three alloys. (a–c) Dislocation structure: Here, arrows mark dislocation lines. Dislocation type is marked by the color of the arrows, from red (pure screw) to blue (pure edge). The interface between FCC and amorphous regions are marked by a gray mesh. Amorphous regions are defined using low local symmetry as defined in the text; (d–f) Atomic displacements: Coloring of atoms indicate their displacement in the cell during a strain increment of unity. Color scale ranges from no displacement (blue) to a displacement of 1 nm (yellow). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

structure within the computational cell at a strain of 40 is presented. Dislocations do interact, however, with the matrix/precipitate interface and cause interfacial rearrangements and local chemical mixing, as seen in Fig. 1(d–e). Such localized atomic motion is conveniently quantified by computing the mean square relative displacements of atom pairs (MSRD), σ_{XY}^2 [29].

$$\sigma_{XY}^2(t_1, t_2) = \frac{\left(\sum_{i=1}^{N_X} \sum_{j=1}^{Z_Y(i)} \left((R_{ij})_{t_2} - (R_{ij})_{t_1} \right)^2 \right)}{(1 + \delta_{XY}) \sum_{i=1}^{N_X} Z_Y(i)} \quad (1)$$

where $(R_{ij})_{t_1}$ is the vector connecting an atom i to a nearest neighbor atom j at time t_1 ; $(R_{ij})_{t_2}$ is that vector at a later time, t_2 ; N_X is the number of atoms of type X , Z_Y is the number of nearest neighbor XY pairs, and $\delta_{XY} = 1$ if $X = Y$, and 0 otherwise. In Fig. 2, MSRDs are computed over a time interval $\Delta t = t_2 - t_1$ corresponding to one deformation cycle ($\Delta \epsilon \sim 0.6$) and plotted as a function of strain. At small strains, $\sigma_{CuCu}^2(t_1 = 0, \Delta t)$ and $\sigma_{QQ}^2(t_1 = 0, \Delta t)$ provide a measure of the rates of shear-induced mixing in FCC Cu and in BCC Q (= Nb, V, Ta) respectively, while $\sigma_{CuQ}^2(t_1 = 0, \Delta t)$ measures mixing at the FCC/BCC interface. Fig. 2 illustrates that very few, if any, atomic relocations occur within the BCC phase, but that relative atomic motion between Cu and Q is extensive at the matrix/precipitate interface. The very large initial value of σ_{CuQ}^2 is a consequence of dislocations nucleating at the precipitate/matrix interface, as seen in Fig. 1(a), before the dislocation density builds up and other nucleation sites become available.

The phase evolutions of these three alloys at these early stages of shear deformation are shown in Fig. 3. Here it is observed that an amorphous interfacial phase begins to develop at the interface with a steep gradient in alloy concentration. It is also observed in these figures that Nb, V, and Ta atoms are transferred across the interface region and into the Cu matrix, where they are carried away from

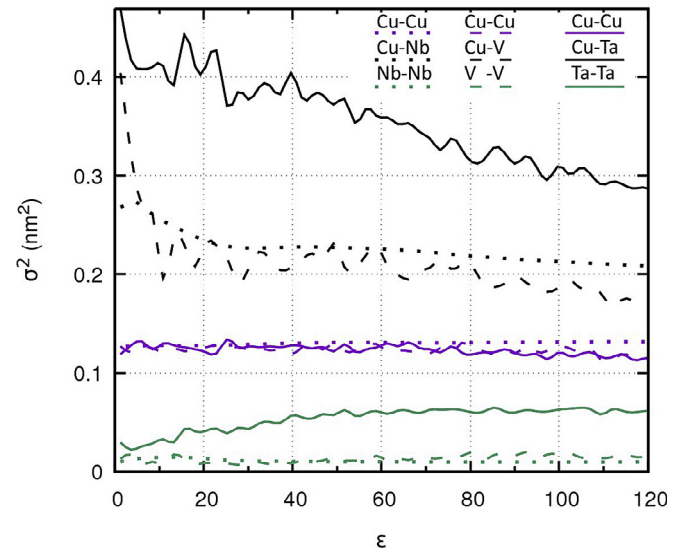


Fig. 2. Mean square relative displacements of atoms (see eq. (1)) plotted as a function of strain for Cu-Cu pairs: σ_{CuCu}^2 (black lines); Cu-Q pairs: σ_{CuQ}^2 (magenta lines); and Q-Q pairs: σ_{QQ}^2 (green lines), where Q = Nb, V or Ta. Alloy system is indicated by type of line: Cu-Nb (dotted), Cu-V (dashed) and Cu-Ta (solid). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the precipitate by dislocation glide. We point that this atomic transport mechanism does not imply that solute atoms are dragged by the dislocations. More simply, if a dislocation passes between the precipitate and the solute atom, the two bodies are displaced relative to each other. After several such events, the solute and precipitate can be well separated [8,29,30]. Cu, on the other hand, does not mix into the BCC precipitates. This initial stage of mixing was partially described for Cu-Nb and Cu-V [30], previously.

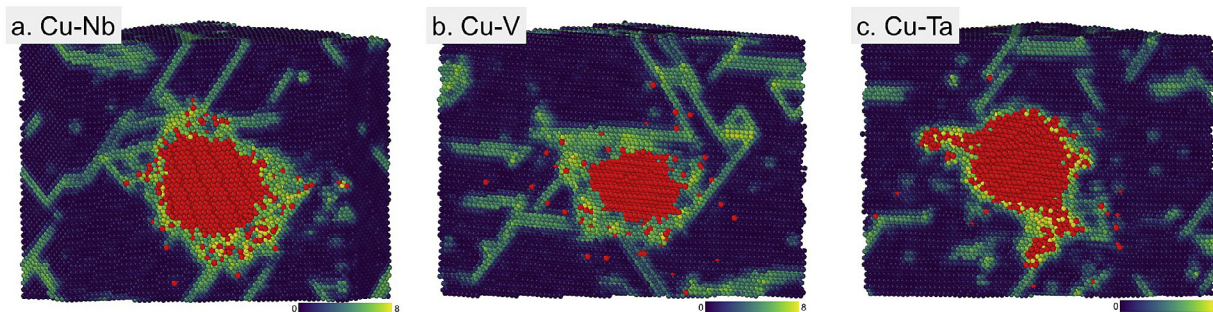


Fig. 3. a–c Snapshots of a (110) cut through the simulated system at a strain of 40, as in Fig. 1. Coloring is indicative of centro-symmetry, ranging from blue, marking atoms at perfect crystalline positions, to yellow, indicative of low symmetry. Non-Cu atoms are displayed with large red symbols. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

In addition to the amorphous shells surrounding the BCC precipitates during this phase of microstructural evolution, small amorphous clusters enriched in solute are found in the Cu matrix, away from the BCC precipitate. For Cu-Ta, some of these clusters have a nearly pure Ta core, ~ 1 nm in size. We traced the trajectories of some of the Ta atoms found in these cores, back to their initial locations in the BCC precipitate, to determine how the clusters form. As shown in Fig. 4(a) for three representative, nearest-neighbor Ta atoms found in one of these clusters, these atoms were not in close proximity in the BCC precipitate at the outset, and their trajectories plotted in Fig. 4(b) led them to the cluster over very different pathways. The zigzag paths illustrate the atomic transport process mediated by dislocation glide as described above. These histories thus show that the Ta clusters are not formed by fracture of the Ta precipitate, but rather by the emission of single Ta atoms into the Cu matrix, where they are transported by dislocation glide and later cluster. These amorphous clusters, moreover, form and annihilate with increasing strain, although we did not investigate the dynamics of this process, owing to their small numbers. A similar formation of such *satellite* clusters, notably, was reported in Cu-W [22] and other immiscible alloy systems [31] during low temperature ion irradiation, again suggesting a commonality between these two forcing conditions.

The amorphous interfacial layers forming at the periphery of the BCC precipitate broaden with continued strain. For several of the Cu-Nb alloys the shearing was continued until the system approached a steady state; this is illustrated in Fig. 5(a) for the 3 at.% Nb alloy. Here it is observed that the initially spherical BCC Nb

precipitate has evolved into an irregularly shaped, small amorphous precipitate, containing $\sim 3 \times 10^4$ atoms and a Nb concentration of ~ 10 at.%. The crystalline phase contains 1 at.% Nb. Following the evolution yet further, shows that the precipitate continually changes its shape, but not its average size and composition, see Fig. 5(b). This evolution is tracked quantitatively in Fig. 6, where the size and composition of the amorphous precipitate are plotted as functions of strain. Initially, the BCC precipitate transforms to an amorphous phase at the precipitate/matrix interface, as described above, and it grows in size. After a strain of $\varepsilon \sim 400$, the amorphous precipitate reaches its maximum size, with an average Nb concentration of ~ 10 at.%. At this level of strain, the system is approaching steady state, although Nb has not yet completely homogenized throughout the crystalline Cu phase, and additional Nb leaves the amorphous region. In response to this loss of Nb, the precipitate begins to shrink as it maintains an average composition at ~ 10 at.% Nb. We discuss this minimum required composition for the amorphous phase in more detail in section 3.3. Lastly, we consider the switching of atoms between the amorphous phase and the crystalline phase to provide an additional measure of the dynamic fluctuations occurring in the steady state. This is illustrated in Fig. 6(b). Here, the atoms in the amorphous phase in the Cu-3 at.% Nb alloy (the same as in Fig. 6(a)) are identified at a specific level of strain, taken here to be $\varepsilon = 400$, which we choose to be the reference state. The amorphous phase has the largest volume at this level of strain. We then identify the atoms in the amorphous phase at all other levels of strain, and plot as a function of strain, the fraction of these atoms that are also in the amorphous phase in the

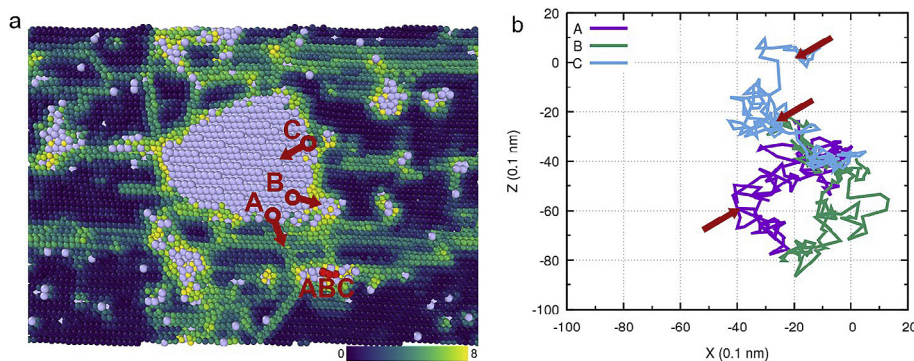


Fig. 4. a. Snapshot of a (110) cut indicating locations of the three specific Ta atoms before deformation (marked as A, B, and C with open circles), and after deformation to a strain of 40 (marked as A, B, and C with closed circles). The initial locations of the three Ta atoms are projections onto the displayed cut. Shading is indicative of the centro-symmetry parameter with non-crystalline Cu atoms colored yellow (see color bar). Ta atoms are marked with larger light blue spheres. b. Projections of the trajectories of the three Ta atoms during the accumulation of strain from 0 to 40 in steps of $\Delta\varepsilon = 0.6$. The trajectories are presented on (010) cut, spanning only the region in which the atoms moved. The final positions of the three atoms are near $x = 0$ and $z = -4$ nm; their initial locations are marked by arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

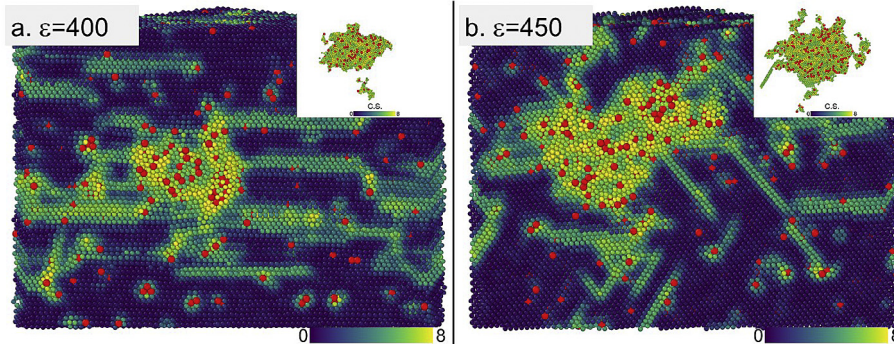


Fig. 5. Snapshots of (110) cuts through a Cu 3 at.% Nb system at strains of 400 and 450 (i.e., after reaching steady state). Coloring in accordance with local symmetry, with blue indicating FCC symmetry and yellow indicating non-crystalline atoms. The small inset for each picture shows locations of the non-crystalline atoms with all crystalline atoms removed. Nb atoms are indicated by large red spheres. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

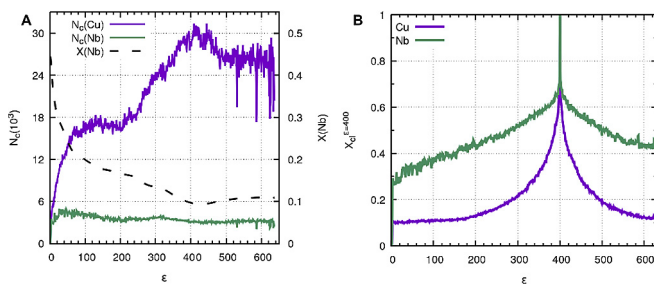


Fig. 6. a. Size and composition of the Nb-rich non-crystalline precipitate for the Cu-3% at. Nb alloy. Up to a strain of 200 the non-crystalline region includes interfacial regions. After a strain of 400 the system reaches steady state with a fully amorphous precipitate and a FCC solid solution with a concentration of approximately 1 at.% Nb. b. Fraction of the Cu and Nb atoms within the amorphous phase at strain ϵ that are also part of the amorphous cluster at $\epsilon = 400$.

reference state. Clearly many atoms switch between the two phases; note particularly that this true even after the system reaches steady state, demonstrating the dynamic nature of the steady state.

3.2. Dependence on initial state

The influence of the initial state of the alloy system on the steady-state microstructure was examined over the full range of presented concentrations, and is demonstrated using a Cu-2 at.% Nb alloy. In one case the system was initiated by embedding a pure BCC Nb spherical precipitate in a pure FCC Cu matrix, while in the other, by creating a random alloy. The results are reported in Fig. 7. Here, a short-range order parameter, as defined in Ref. [16],

$$\Omega = \frac{\left(\left(\frac{c}{1-c} \right) N_{Nb}^{Cu} + \left(\frac{1-c}{c} \right) N_{Cu}^{Nb} \right)}{\left(N_{Cu}^{Cu} + N_{Nb}^{Nb} \right)} - 1$$

is plotted as a function of strain for the two systems, one starting at $\Omega \sim -1$ (phase separated) and the other at $\Omega \sim 0$ (random solution); c is the concentration of Nb, and N_A^B is the average number of B atoms that are nearest neighbors of A atoms. After a strain of ~ 125 , the short range order parameter for the two systems merge at $\Omega \sim -0.07$. This value appears consistent with the degree of phase separation visually observed (i.e., a value lower than 0) and it remains constant thereafter. Also shown in Fig. 7 are cross sectional slices through the precipitates, distinguishing the amorphous phase by the local symmetry relative to a FCC structure. While the shapes of the two amorphous precipitates are somewhat different, this difference is typical of shape variations that occur in steady state under shear, as discussed above. The volumes of the two precipitates, however, are nearly the same, and the average composition of the precipitates is again ~ 10 at.% Nb.

It can be noticed in Fig. 7 that the initially homogeneous Cu-Nb system approaches steady state far sooner than the phase-separated system. This occurs simply because the kinetic pathways leading to steady state are very different; even the mixing mechanisms for the initial states are different. For example, the system starting with a BCC Nb precipitates requires transferring the Nb atoms from the crystalline Nb phase into the amorphous phase, and this is slow relative to the transport of Nb atoms in the amorphous phase or the crystalline Cu phase. We in fact take advantage of this faster kinetics in the initially homogenous alloy in the remainder of this work to consider the effects of alloy composition on the final microstructure, but with the tacit assumption that all of

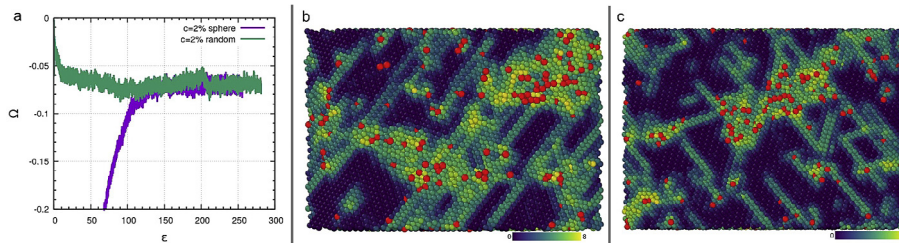


Fig. 7. (a) Short range order parameter as a function of strain for Cu-2 at.% Nb initiated as a random solution (green) or as a solid sphere (magenta). (b) and (c): (110) cuts through systems initiated as a sphere (b) or as a random solution (c) at steady state. Coloring in accordance with local symmetry, with blue indicating FCC symmetry and yellow indicating non-crystalline atoms. Nb atoms are indicated by large red spheres. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

our systems, not just Cu-Nb, show the same behavior in steady state, independent of starting condition.

3.3. Dependence of the steady state microstructure on alloy composition: the lever rule during SPD

Fig. 8(a–d) plot the fraction of the amorphous phase, f_a , and the molar fraction of solute Q (= Nb, V, and Ta) in the amorphous phase, as functions of average system concentration. Fig. 8(a) shows that the three alloys begin to become two-phase, crystalline and amorphous, at very low solute concentrations, ~0.3 at.% Ta, ~1 at.% Nb, or ~5 at.% V, while they become homogeneous amorphous phases above ~8 at.% V, 9 at.% Nb, or 24 at.% Ta. Between these two limits, the steady state microstructure is two-phase, a highly Cu-rich crystalline phase and a solute enriched amorphous phase. As shown in Fig. 8(b–d), the compositions of the amorphous phases in this two-phase region remain fixed, independent of average alloy composition. For the crystalline phases, the compositions also remain fixed, 5 ± 1 at.%, 1 ± 0.5 at.%, and 0.5 ± 2 at.% for V, Nb, and Ta, respectively (data not shown). The lever rule is thus obeyed in these alloys during lower temperature SPD. The high solubility of V in the crystalline phase and very low solubility of Ta are consistent with their relative solubilities in Cu in equilibrium. For all three systems, we wish to emphasize for later discussion that the precipitates are singly connected, and their sizes are proportional to the alloy concentration and the size of the computational box.

The structure of the Cu-Ta system is somewhat different than the other two alloys. First it remains two phase up to concentrations of ~24 at.%. The reason for the high average concentration of Ta appears to stem from the fact that the amorphous phase is itself two phase, containing both Cu rich and Ta rich amorphous phases. While it is difficult to determine the exact concentrations of these two amorphous regions owing to their small length scale and asymmetric shape (to be discussed below), they are approximately ~10 at.% Ta and nearly pure Ta.

For completeness in describing these simulations, we note that in constructing these phase diagrams, we started each simulation with a homogeneous alloy. For most alloy concentrations above the

solubility limit in the crystalline phase, the alloys spontaneously formed co-existing crystalline and amorphous phases during shearing, as described. For alloys with concentrations approaching the phase boundary for complete amorphization, however, the system initially underwent amorphization globally, before solute partitioning could occur. In these cases a small crystalline nucleus, approximately 1 nm in diameter, was embedded in the amorphous phase to nucleate the crystalline phase. For Cu-Nb this occurred for concentrations greater than 5 at. %Nb. The addition of such crystalline nuclei to the amorphous regions at lower concentrations, or to the amorphous phase at concentrations just above the upper phase boundary, had no effect on the observed steady states.

3.4. Comparison of liquid Cu-Ta with amorphous Cu-Ta under shear deformation: effective temperature model

The observation that the lever rule is obeyed during low-temperature shearing, similar to equilibrium two-phase alloys, is perhaps suggestive that shearing can be associated with an effective temperature. We thus compared the microstructure of the Cu-Ta system in the liquid state with that formed at low temperature during shearing. We chose the Cu-Ta systems for this part of the study since past work by Frolov and Mishin had revealed that Cu-Ta alloys containing 20–50 at.% Ta formed thermodynamically stable colloidal structures between 2000 K and 3500 K, with nanometer-size Ta particles suspended in liquid Cu [32]. We thus selected alloys containing 20 at.% Ta and 25 at.% Ta alloys for shear deformation, as these compositions are respectively just below, and just above the phase boundary for complete amorphization during low-temperature shearing. Plotted in Fig. 9 is the short-range order parameter, Ω , defined by eq. (1), as a function of strain. Also shown are values of Ω obtained during equilibration of these alloys at various temperatures, in absence of shearing. These results suggest, from simply pairing similar values of Ω [see e.g., [16]], that low-temperature shearing of Cu-Ta at $1 \times 10^8 \text{ s}^{-1}$ corresponds to an effective temperature of ~2000 K. Examination of the shear-induced and equilibrium structures in Fig. 10, moreover, provides additional insight. First we note that in addition to the similar short

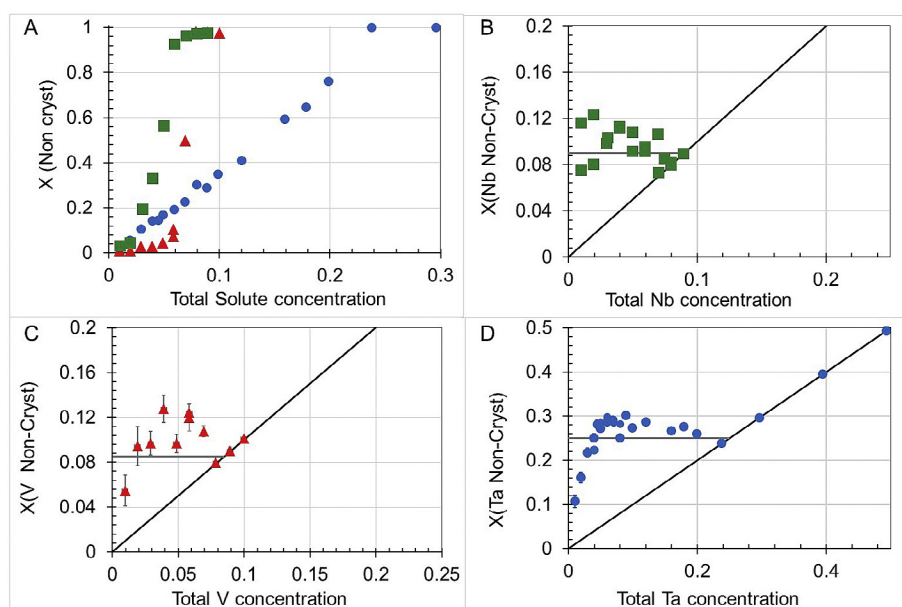


Fig. 8. a. Fraction of atoms, $X(\text{Non-cryst})$, within the amorphous phase at steady state, plotted as a function of solute concentration for Cu-Nb (green squares), Cu-Ta (blue circles) and Cu-V (red triangles). b, c, d. Molar fraction of Nb, V and Ta within the non-crystalline regions respectively. Lines with slope = 1 represent the average concentrations in the two-phase regions at steady state. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

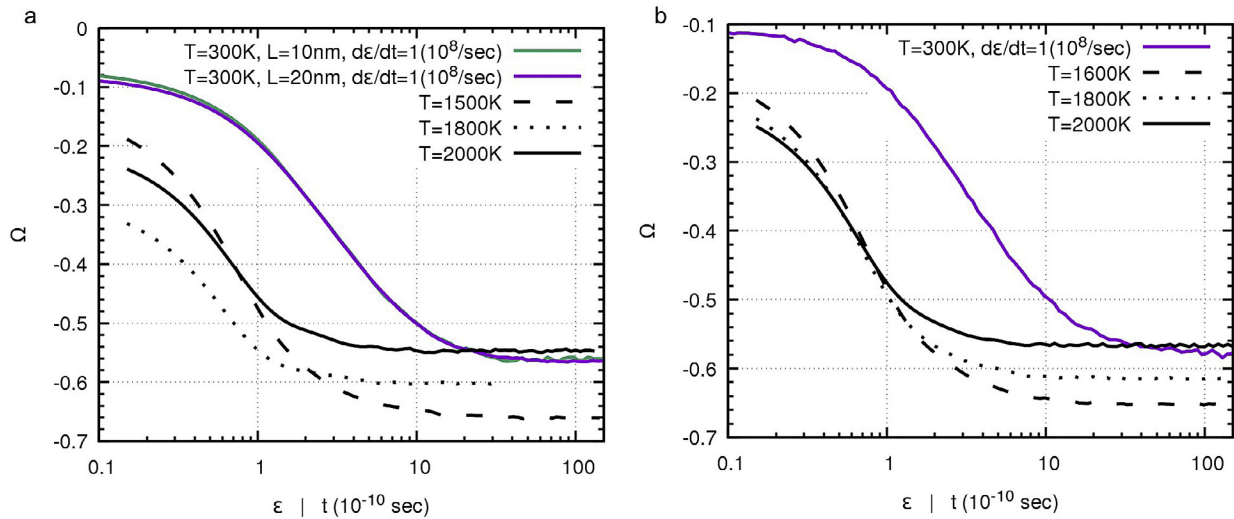


Fig. 9. Short range order parameter versus strain, during shearing, or time, during annealing, for Cu-20 at.% Ta (a) and Cu-25 at.% Ta (b). SPD samples were simulated at 300 K with a strain rate of 10^8 (1/sec). Order parameters for systems annealed at fixed temperatures in absence of shear are plotted as a function of time (in units of 0.1 nsec). Annealing temperatures are indicated in the legend. For the 20 at.% Ta alloy results from simulation box sizes of $(10\text{ nm})^3$ and $(20\text{ nm})^3$ are shown to demonstrate system size convergence.

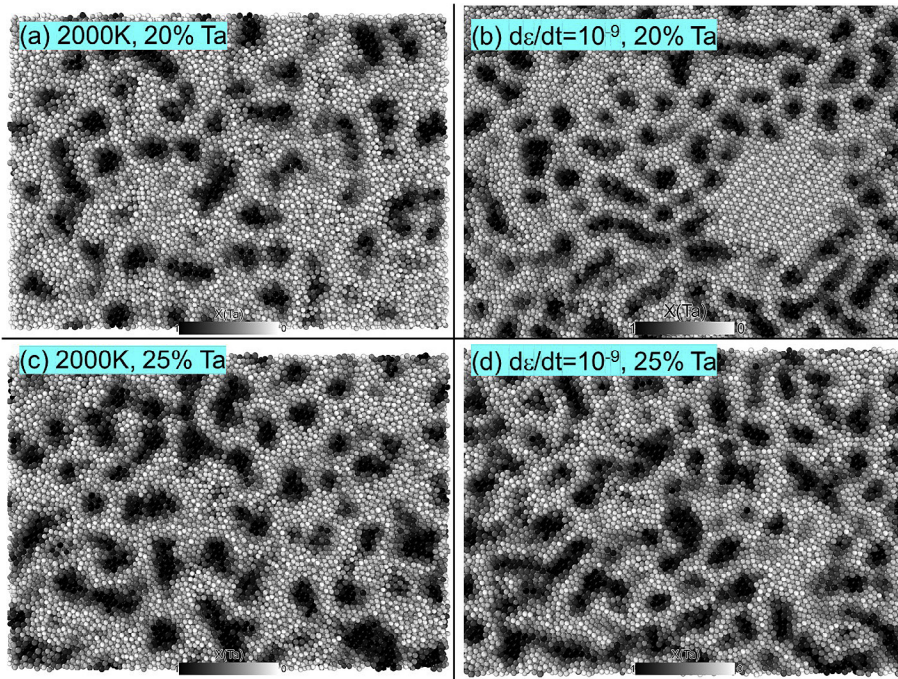


Fig. 10. Snapshots of Cu-20 at.% Ta and Cu-25 at.% Ta showing similarity between the non-sheared sample equilibrated at 2000 K and the steady state SPD sample at 300 K. Gray scale represents the local concentration of Ta atoms, pure Ta, black; pure Cu, white.

range order parameters, the length scales of the phase separation for the high-temperature equilibrium alloys are remarkably similar to those of the sheared alloys. We quantified these length scales by determining the size distribution of the Ta-rich clusters. This was done by identifying contiguous clusters of Ta atoms which had a nearest-neighbor Ta concentration higher than 90%. As shown in Fig. 11, these distributions are, indeed, very similar. The main difference in the structure between the equilibrium and sheared alloys is found in the 20 at.% Ta where the sheared alloy has formed a Cu-rich crystalline phase. This, of course, cannot occur in the equilibrium structure as the temperature is above the melting point of Cu. As a final point, we note that on increasing the strain rate a

factor of two, to $2 \times 10^8\text{ s}^{-1}$, results in an increase in the effective temperature by $\sim 200\text{ K}$. We tentatively attribute this increase in effective temperature to the larger applied stresses, which increase from 20 to 25 kbar on increasing the strain rates; with these higher stresses the alloy system can be driven to higher states of free energy, see e.g., refs. [16,36].

4. Discussion

The primary findings of this work are (i) Cu-Nb, Cu-V and Cu-Ta form two phase alloys during low-temperature, steady-state, shear deformation over an extended range of alloy compositions, with all

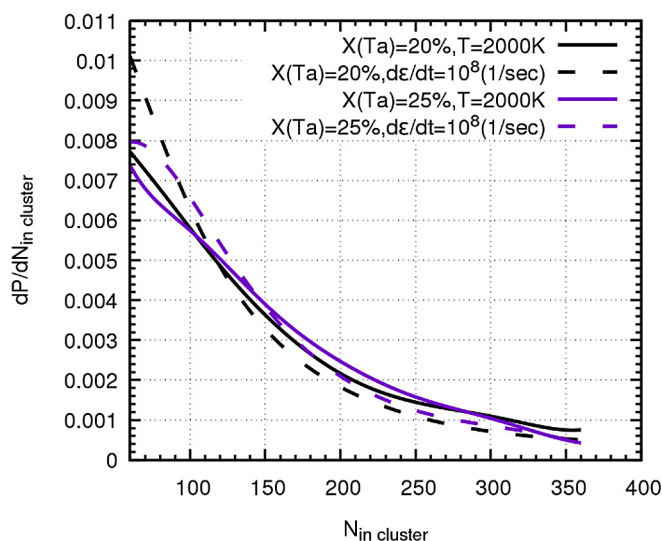


Fig. 11. Cluster size distributions plotting the probability densities of Ta-rich clusters of a specific number of atoms, $dP/dN_{in\ cluster}$, as a function of cluster size, $N_{in\ cluster}$ for Cu-20 at.% Ta (black lines) or 25 at.% Ta (magenta lines) at steady state. Distributions for shearing at 300 K marked are plotted with dashed lines, and systems annealed at 2000 K under zero stress conditions are plotted with solid lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

systems following a lever rule reminiscent of two-phase alloys in equilibrium. At alloy compositions below the saturation limit for solute in Cu, the systems form single, solid-solution, FCC phases, while above a second critical concentration limit, they form amorphous phases. For Cu-Ta, the amorphous phase separates in to Cu rich and Cu poor regions; (ii) for the Cu-Nb system, steady states are independent of their initial microstructure. Cu-V and Cu-Ta were not completely tested for this invariance, but the initial trajectories of their microstructures suggest similar behavior; (iii) the strain required to reach steady state, on the other hand, depends on the initial state of the system; (iv) chemical short range order and phase partitioning in amorphous Cu-Ta under shear shows strong similarities to that in Cu-Ta above the melting temperature.

We first compare these findings with recent experiments on the Cu-Nb system. Homogeneous Cu-Nb alloys in these experiments were prepared by magnetron sputter deposition with Nb concentrations varying from ~1 at.% Nb to ~10 at.% Nb. They were then subjected to high pressure torsion (HPT) at ~78 °C to a strain of ~2500. At 78 °C, thermally activated migration of vacancies is negligible in Cu. The results showed that (i) below ~1 at.% Nb, the system formed a single phase FCC structure, while at concentrations between 2 at.% and 10 at.% Nb, the system formed two-phase alloys, a FCC phase containing ~1.5 at.% Nb and a BCC, nearly pure, Nb phase. This work also showed that if these alloy specimens were first annealed to high temperatures to form Nb precipitates with diameters ~120 nm, subsequent HPT deformation at ~78 °C led to refinement of the precipitate size and microstructures that approached those formed without the preliminary heat treatment. The current MD simulations thus agree very well, at least qualitatively, with these experiments.

Two aspects of the experiment, however, seemingly differ from the simulations: the precipitate particles are BCC rather than amorphous, and the size of these Nb-rich precipitates were ~20 nm in diameter and thus significantly larger than those found here. These differences can be due to a number of factors, e.g., inadequacies of the interatomic potentials and vast differences in strain rates and strain paths, but we focus here on the limited size of

the computational cell. For all the systems studied here, see e.g., inset of Fig. 5, the precipitated phase was comprised of a single (amorphous) precipitate. Since the volume fraction of the amorphous phase in the system is determined by the lever rule, the largest possible size of this one precipitate is determined by the phase boundaries of the two co-existing phases and the size of the computational cell. Consequently, the size of the precipitate scaled closely with the size of the computational cell for a given concentration. Thus, the size of the Nb precipitates in the experiments cannot be directly compared with the sizes of the precipitates in the simulations. For these simulations to capture large sizes of precipitates found in the experiment, therefore, the size of the computational cell would need to be increased a factor of ~100, and hence not presently feasible. The limited size of the computational cell, and hence precipitate size, may possibly also explain why the precipitates are amorphous. Note, as the precipitate size becomes small, the interfacial free energy becomes an increasingly important contribution to the total free energy of the precipitate. The interfacial free energy of a crystalline, BCC Nb-rich particle in Cu, moreover, is likely to be larger than that of a Cu-10 at.% Nb amorphous particle, owing to its larger number of Cu–Nb bonds. Although we did not explore this point further, we note that small amorphous regions in Cu-Nb subjected to severe plastic deformation have indeed been reported in the literature [33–35].

There has been some question in the literature, whether driven alloy systems, either by irradiation or severe plastic deformation are ergodic, or more specifically here, whether the steady state of the system is independent of its initial state. While the three systems investigated here appear so, Cordero and Schuh have suggested that elements with a large difference in shear strength may behave otherwise [17]. This conclusion was based on KMC simulations and evaluations of past experiments. The simulations revealed that initially phase-separated alloys remained phase-separated, while the same alloys remained homogeneous if initially homogeneous. While the present simulations cannot answer this question more broadly than stating that the steady state in Cu-Nb system during low-temperature shear is independent of the initial state, and that Cu-V and Cu-Ta appear to behave similarly, they do offer additional insights. The KMC model adopted by Cordero and Schuh employed a rigid lattice model, whereby shear was achieved by shifting atomic planes, analogous to ref. [8]. The present MD simulations show, however, that incoherent interfaces and lattice dislocations play a key role in the shear mixing process. In the three systems investigated here, for example, mixing of the initially phase-separated systems began at an incoherent interface, which soon developed an amorphous phase. Similarly, the Ta rich clusters that nucleated near the initial BCC Ta precipitates were not coherent with the Cu lattice. These reactions are missed in rigid lattice models. The MD simulations also show that the strain required to reach steady state depends on the initial state, as well as on the size of the computational cell. It appears therefore, that there is simply no easy method for predicting the time and length scales required to adequately test for steady states, or more generally, for ergodic behavior, either by simulation or by experiment.

Lastly we comment briefly on the question of corresponding states, i.e., can an effective temperature be assigned to low-temperature shear mixing. This description was first addressed by Martin [36] for irradiated materials, but later broadened to shear deformation [37]. The key feature in this model is a competition between forced mixing, i.e., recoil of atoms, disordering the alloy, and thermally activated diffusion returning the system to equilibrium. While this model considered systems at elevated temperatures, Lund and Schuh [16], extended Martin's work in molecular statics simulations to include low temperature

shearing, where thermally activated diffusion mechanisms are suppressed. As pointed out above, this model appears to work well for describing the fully amorphous Cu-Ta under low temperature shear, i.e., when the Ta concentration exceeds 24 at.%. In this case, the extended effective temperature model captures both the short range order, Ω , see Fig. 10(a), and the size and the length scale of Ta rich clusters, see Fig. 10(b). Two points are worth mentioning. First, the system considered by Lund and Schuh, similar to 25 at.% Ta system presented here, is fully amorphous, and second the short range order parameter and the length scale of the Ta clusters found at elevated temperatures derive from equilibrium thermodynamic origins, alone; note the interatomic potential for Cu-Ta yields a curvature dependent, negative interface energy which yields stable Ta nanoclusters in liquid Cu [32]. While a discussion of the importance these features goes beyond the scope of this work, we point out that neither of these features is true for the 20 at.% Ta system, where a large crystalline, Cu-rich phase is observed. This crystalline phase introduces not only a non-amorphous regions, but also a new length scale into the problem, viz, the glide distance of dislocations. As shown by Odunuga et al. [29], the effective relative diffusion coefficients of pairs of atoms due to dislocation glide are linearly proportional to their distance apart. Consequently, the forced mixing term depends on the domain size of the crystalline phase, and this length scale is not of thermodynamic origin. The competition between forced mixing and relaxation, therefore, can be far different in amorphous and crystalline systems. This is important since it has been shown previously, for both irradiation [22,38] and shearing [39], that this competition can lead to well-defined length scales in the phase separation, i.e., compositional patterning. In this regard, the experiments on low-temperature SPD of Cu-Nb alluded to above [20,35], reveal a steady-state microstructure comprised of 20 nm Nb BCC precipitates in a Cu matrix. This, however, cannot represent an equilibrium structure and therefore it cannot have a corresponding state at any temperature.

5. Conclusions

This work has shown that Cu-Nb, Cu-V and Cu-Ta form two-phase alloys during low-temperature, steady-state, shear deformation over an extended range of alloy compositions, a Cu-rich crystalline phase and a solute enriched amorphous phase. The three alloy systems all follow a lever rule reminiscent of two-phase alloys in equilibrium, suggesting that severe plastic deformation can possibly be described by an effective temperature model. Detailed comparison of the microstructures formed in Cu-Ta during low temperature shear with those obtained by equilibration at high temperatures showed that the short-range order of the system equilibrated at 2000 K is quite similar to the sheared system, at least in the amorphous state. These findings were rationalized by considering the detailed mechanisms underlying shear mixing at low temperatures.

The simulations also revealed that the Cu-Nb system approaches a steady-state microstructure under low temperature shear deformation independent of the initial microstructure, completely phase separated, or homogeneous. The strain required to reach steady state, however, was much different for the two starting conditions. In all three alloys, the amorphous phase in the two-phase region was comprised of a single precipitate, with size scaling with the size of the computational cell. The simulations were shown to agree qualitatively with experimental observations, but due to the small size of the computational cell, detailed quantitative comparison was not possible.

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